Operating System (OS)

1. What is an Operating System (OS)?

An Operating System (OS) is system software that acts as an interface between the user and computer hardware.

It manages all hardware and software resources and allows users to interact with a computer.

Simple Definition:

An Operating System is software that controls computer hardware, manages applications, and provides an environment for users to perform different tasks.

Examples:

- Microsoft Windows 10 / 11
- Linux
- macOS
- Android
- iOS
- ChromeOS

Introduction to Windows 11 - SCS Computing Facilities - Carnegie Mellon University

Ubuntu 20.04 LTS arrives

File:MacOS-Monterey-21C52 desktop.png - BetaWiki

Real-life example

Think of a computer as a company:

- Hardware = Employees and machines
- Applications = Different departments
- User = Customer
- Operating System = Manager

The manager coordinates all activities. Similarly, an OS manages all computer resources.

2. Why Do We Need an Operating System?

Without an operating system, using a computer would be very difficult.

An OS is required for the following reasons:

1. Hardware Management: Controls CPU, RAM, keyboard, mouse, printer, etc.
2. Memory Management: Allocates and manages RAM for applications.
3. File Management: Creates, stores, deletes, and organizes files and folders.

4. Process Management: Manages running applications and background processes.
5. Device Management: Controls hardware devices through device drivers.
6. Security Management: Provides passwords, user accounts, permissions, and access control.
7. User Interface: Provides GUI and CLI to interact with computers.
8. Networking: Manages internet and network communication.
9. Application Management: Provides an environment to install and run software.
10. Error Handling: Detects and handles system errors.

3. Main Components of an Operating System

Kernel Development Series Phase 0: Orientation & Big Picture - Wasil Zafar

wasilzafar.com

Component	Function
Kernel	Core part of OS; manages hardware and resources
Shell	Interface between user and OS services
File System	Manages files and folders
Device Drivers	Allow OS to communicate with hardware
Process Manager	Manages running processes
Memory Manager	Allocates RAM
Security Manager	Controls user permissions and access

What is a Kernel?

The kernel is the core component of an operating system. It manages CPU, memory, devices, and processes.

Examples:

- Windows NT Kernel
- Linux Kernel
- XNU Kernel (macOS)

What is Shell?

A shell provides an interface through which users interact with the operating system.

Two major types:

- GUI (Graphical User Interface): Windows Desktop, Ubuntu Desktop.
- CLI (Command Line Interface): CMD, PowerShell, Linux Terminal.

4. Types of Operating Systems

There are several types of operating systems based on their working methods.

4.1 Batch Operating System

A Batch OS executes multiple jobs in groups (batches) without requiring continuous user interaction.

Example: Early IBM mainframe systems.

Applications: Payroll processing, banking transactions, bulk data processing.

4.2 Time-Sharing Operating System

A Time-Sharing OS allows multiple users to use computer resources simultaneously by allocating CPU time to each user or task.

Example: UNIX, Linux.

Applications: Multi-user servers and university computer systems.

4.3 Multitasking Operating System

A Multitasking OS allows users to run multiple applications at the same time.

Examples:

- Windows 11
- macOS
- Linux

Practical example: Running Chrome, MS Word, and VLC simultaneously.

4.4 Multiprocessing Operating System

A Multiprocessing OS uses two or more CPU cores or processors to execute multiple processes.

Example: Windows, Linux, macOS.

Advantages: Better performance, parallel processing, and efficient resource utilization.

4.5 Multi-User Operating System

A Multi-User OS allows multiple users to access the same computer system or server.

Examples: Linux, UNIX, Windows Server.

Applications: Company servers, hosting servers, and university labs.

4.6 Real-Time Operating System (RTOS)

An RTOS processes data and responds to events within a specified time limit.

Types:

- Hard Real-Time OS: Missing a deadline can cause system failure.
- Soft Real-Time OS: Small delays are acceptable.

Examples: FreeRTOS, VxWorks.

Applications: Medical equipment, robotics, industrial automation, and aircraft control systems.

4.7 Distributed Operating System

A Distributed OS manages multiple connected computers and makes them work together as a coordinated system.

Applications: Distributed computing and high-performance computing.

4.8 Network Operating System (NOS)

A Network OS manages network resources, users, permissions, shared files, printers, and centralized services.

Examples:

- Windows Server
- Linux Server
- UNIX

Applications: Offices, schools, organizations, and data centers.

4.9 Embedded Operating System

An Embedded OS is designed for dedicated devices that perform specific functions.

Examples: Embedded Linux, FreeRTOS.

Applications: Smart TVs, routers, smart watches, and IoT devices.

Types of OS – Quick Revision

Type	**Main purpose**
Batch OS	Executes jobs in batches
Time-Sharing OS	Shares CPU time
Multitasking OS	Runs multiple applications
Multiprocessing OS	Uses multiple processors/cores
Multi-User OS	Supports multiple users
Real-Time OS	Responds within deadlines
Distributed OS	Coordinates multiple computers
Network OS	Manages network resources
Embedded OS	Controls dedicated devices

5. What is Virtualization?

Virtualization is a technology that allows us to create multiple virtual computers on a single physical computer.

A virtual computer behaves like a real computer and can have its own:

- Operating System
- RAM
- Virtual CPU
- Virtual Hard Disk
- Network Adapter
- Applications

Example:

Suppose your physical laptop has Windows 11 installed. You can create a virtual machine and install Ubuntu Linux inside it.

You can use both operating systems without removing your existing Windows installation.

What Is VM: Virtual Machine Definition & How It Works in 2026

monovm.com

5.1 What is a Virtual Machine (VM)?

A Virtual Machine is a software-based computer that runs inside a physical computer.

Example:

Physical Computer (Host Machine)

Windows 11 • 16 GB RAM • 512 GB SSD

Virtualization Software

Oracle VirtualBox

VM 1
Ubuntu Linux

4 GB RAM • 40 GB disk

VM 2
Windows 10

4 GB RAM • 60 GB disk

5.2 What is a Hypervisor?

A Hypervisor is software that creates and manages virtual machines.

There are two main types:

Type 1 – Bare Metal Hypervisor

Runs directly on physical hardware.

Examples:

- VMware ESXi
- Microsoft Hyper-V (bare-metal deployment)
- Xen

Type 2 – Hosted Hypervisor

Runs as an application inside an existing operating system.

Examples:

- Oracle VirtualBox
- VMware Workstation
- VMware Fusion

Bare Metal Hypervisors: A Deep Dive For Everyone

Hypervisors in Cloud Computing | by Afzal Badshah, PhD | Medium

kim.dragon :: [가상화] 가상화란 무엇인가?

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5.3 Benefits of Virtualization

1. Run multiple operating systems on one computer.
2. Practice Linux and Windows Server.
3. Learn ethical hacking in isolated lab environments.
4. Test software without affecting the main OS.
5. Practice networking and server configuration.
6. Create snapshots and restore previous states.
7. Reduce hardware costs.

6. Practical: How to Create a Virtual Machine

We will use Oracle VirtualBox to create an Ubuntu Linux virtual machine.

1.9. Using VM Groups

docs.oracle.com

Step 1: Check System Requirements

Before creating a VM, check your laptop or desktop configuration.

Component	Recommended for this lab

Processor	Intel Core i5 / AMD Ryzen 5 or better
RAM	8 GB minimum; 16 GB recommended
Storage	50 GB free space
Virtualization	Intel VT-x / AMD-V enabled
OS	Windows 10 / 11

Step 2: Download VirtualBox

VirtualBox | Unix Tutorial

Oracle VirtualBox

Free virtualization software

Download VirtualBox

1. Open the official website.
2. Download the Windows host installer.
3. Run the installer.
4. Click Next.
5. Accept the default installation options.
6. Click Install.
7. Open VirtualBox.

Step 3: Download Ubuntu ISO File

대학생이 리눅스 우분투를 2년간 써보면서 (Ubuntu 22.04) — 쓰기 전에 생각하자

Ubuntu Linux

Linux operating system

Download Ubuntu ISO

An ISO file is a disk image containing the operating system installation files.

Example filename: ubuntu-24.04-desktop-amd64.iso

Step 4: Create a New Virtual Machine

Virtual Box: Setting Up Linux - LSHTM Omics

lshtm-genomics.github.io

1. Open VirtualBox.
2. Click New.
3. Enter VM Name: Ubuntu-Linux.

4. Select ISO Image: Ubuntu ISO file.
5. Select OS Type: Linux.
6. Select Version: Ubuntu (64-bit).
7. Allocate RAM: 4096 MB (4 GB).
8. Allocate CPU: 2 processors.
9. Create a virtual hard disk.
10. Select VDI format.
11. Choose Dynamically Allocated.
12. Set virtual disk size to 40 GB.
13. Click Finish.

These are example settings for a computer with sufficient resources. Reduce allocation if your host computer has limited RAM or CPU capacity.

Step 5: Start the Virtual Machine

How To Install Ubuntu on Virtualbox

linkedin.com

1. Select Ubuntu-Linux.
2. Click Start.
3. Ubuntu installation screen will appear.
4. Select your language.
5. Click Install Ubuntu.

Step 6: Install Ubuntu Operating System

How To Dual Boot Linux and Windows on any PC | Tom's Hardware

Install Ubuntu Desktop | Ubuntu

Install Ubuntu Desktop - Tutorials - Ubuntu Community Hub

Follow these steps:

1. Select language.
2. Select keyboard layout.
3. Choose Normal Installation or Default Installation.
4. Select the virtual disk installation option.
5. Configure your username and password.
6. Select your time zone.
7. Click Install.

8. Wait for installation to complete.
9. Restart the virtual machine when prompted.
10. Log in using your created credentials.

Important: The erase-disk option here applies to the VM's virtual hard disk, not your physical Windows drive. Make sure you are installing inside the VirtualBox VM.

Step 7: Verify Ubuntu Installation

After restarting, log in to Ubuntu.

Open Terminal and execute:

```
# Check operating system
cat /etc/os-release

# Check kernel version
uname -r

# Check IP address
ip addr

# Check available RAM
free -h

# Check disk space
df -h
```

If these commands return the expected system information, your Ubuntu virtual machine is working.

7. Types of Virtual Machine Network Modes

Networking is very important for Cybersecurity and Networking students.

VirtualBox provides different networking modes:

Mode	Explanation
NAT	VM accesses the internet through the host
Bridged Adapter	VM appears as a separate device on the physical network
Host-Only	Communication between host and VM

Internal Network Communication between selected VMs only

Virtual Box Networking - KodeKloud Notes

notes.kodekloud.com

Practical activities:

- Connect Ubuntu VM to the internet.
- Check the IP address.
- Ping the host computer.
- Connect two virtual machines.
- Share files between VMs.
- Practice network scanning in an isolated lab.

8. How to Install Windows Operating System on a Physical Computer

Now let's understand the actual OS installation process using a bootable USB drive.

How to Install Windows 11 A Complete Guide

blogs.digitalhubkey.com

Requirements

- Windows 10/11 ISO file.
- 8 GB or larger USB drive.
- Computer or laptop.
- Internet connection.
- Backup of important data.

Step 1: Download Windows ISO

Official Microsoft download:

Download Windows 11

Step 2: Create a Bootable USB

Download Rufus:

Download Rufus

Cara Menggunakan Rufus untuk Bootable Flashdisk | kumparan.com

kumparan.com

1. Insert USB drive.
2. Open Rufus.

3. Select your USB device.
4. Select Windows ISO.
5. Select partition scheme (usually GPT for modern UEFI computers).
6. Click Start.
7. Wait until the bootable USB is ready.

Warning

Creating a bootable USB will erase its existing data.

Step 3: Boot from USB

1. Restart your computer.
2. Open the Boot Menu.
3. Select your USB drive.
4. Windows Setup will open.

Common boot menu keys:

Brand	Boot Menu Key
Dell	F12
HP	F9 or Esc
Lenovo	F12 or Fn + F12
Acer	F12
ASUS	Esc or F8
MSI	F11

Keys can vary by model.

Step 4: Windows Installation

Der neue Einrichtungsbildschirm in Win11 ist endlich neu

How to Create and Use Windows Installation Media [Windows 11/10] | Syncios

How can I install Windows 11 24H2 Pro without a Microsoft account? - Super User

1. Select language.
2. Select time and currency format.
3. Select keyboard layout.
4. Click Install Now.
5. Enter product key or select the appropriate option to continue without one.

6. Select Windows edition if prompted.
7. Accept license terms.
8. Select Custom Installation for a clean installation.
9. Select the correct target partition.
10. Click Next.
11. Wait for Windows installation.
12. Computer will restart.
13. Complete initial Windows setup.
14. Install drivers and Windows updates.
15. Activate Windows with a valid license.

Important: Selecting, formatting, or deleting partitions can permanently erase data. Always verify the correct disk and take a backup before installing.

9. Difference Between Physical OS Installation and Virtual OS Installation

Feature	**Physical Installation**	**Virtual Installation**
Installation location	Physical HDD/SSD	Virtual hard disk
Hardware	Direct hardware access	Virtual hardware
Performance	Generally higher	Some overhead
Multiple OS	Possible with dual boot	Multiple VMs possible
Risk	Can affect existing data	Usually isolated
Purpose	Daily computer use	Testing and learning
Example	Windows 11 on laptop	Ubuntu in VirtualBox

10. Practical Lab Activities for Students

These activities will help students understand OS installation, virtualization, and networking.

Operating System Practical Lab

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Identify the installed OS and its version.

Check CPU, RAM, and storage configuration.

Identify BIOS and UEFI settings.

Download and identify an ISO file.

Install Oracle VirtualBox.

Create an Ubuntu virtual machine.

Install Ubuntu Linux inside the VM.

Execute basic Linux terminal commands.

Configure NAT and Bridged networking.

Check IP address and test ping.

Create a snapshot and restore the VM.

Install Windows using a bootable USB (demonstration or supervised lab).

Reset practical checklist

11. Interview and Viva Questions

1. What is an Operating System?
2. Why do we need an OS?
3. What is the difference between hardware and software?
4. What is a kernel?
5. What is the difference between GUI and CLI?
6. What is multitasking?
7. What is a real-time operating system?
8. What is a Network Operating System?
9. What is virtualization?
10. What is a Virtual Machine?
11. What is a Hypervisor?
12. What is the difference between Type 1 and Type 2 Hypervisors?
13. What is an ISO file?
14. What is a bootable USB?
15. What is the difference between NAT and Bridged networking?

12. Recommended Learning Roadmap

For students learning Computer Hardware, Networking and Cybersecurity, follow this order:

1. Operating System Fundamentals

Windows, Linux, macOS, OS architecture, kernel and shell.

2. Windows Operating System

Installation, partitioning, drivers, users, permissions and troubleshooting.

3. Linux Operating System

Ubuntu, Kali Linux, terminal commands, file permissions and package management.

4. Virtualization

VirtualBox, VMware, virtual machines, snapshots and networking.

5. Networking

IP addressing, DNS, DHCP, subnetting, routers and switches.

6. Cybersecurity Lab

Kali Linux, Wireshark, Nmap and basic ethical hacking in authorized lab environments.